



AKGEC SKILLS FOUNDATION

Training Program Details

CENTRE NAME- BOSCH AUTOMOTIVE CENTRE OF EXCELLANCE

Training/Course Title:	Industrial Training Program on Race Car Vehicle Design
Target Audience:	B.Tech Students
Fee:	11000/-
Duration:	160 Hrs.
Module	• Introduction to motorsports categories, vehicle types, and FIA technical regulations. • Fundamentals of race vehicle dynamics including load transfer, traction, and tire behaviour. • Suspension geometry, kinematics, and high-speed handling optimization. • Chassis design, structural analysis, torsional rigidity, and crashworthiness. • Powertrain selection, integration, torque curve analysis, and gear ratio tuning. • Drivetrain systems, differential types, torque vectoring, and shaft design. • Brake system engineering, thermal calculations, bias control, and fade resistance. • Aerodynamics fundamentals including wings, diffusers, underbody flow, and CFD simulation. • Motorsports electronics, sensors, DAQ, CAN communication, and telemetry analysis. • Final integrated race car design project including CAD modeling, subsystem validation, and presentation
Outcome:	• Understand the Motorsports Ecosystem and Regulations • Apply Vehicle Dynamics and Suspension Design Principles • Design and Analyze High-Performance Chassis Structures • Integrate Powertrain and Drivetrain Systems for Racing • Engineer Efficient Brake, Steering, and Thermal Systems • Develop Telemetry and Motorsports Electronics Applications • Execute a Complete Race Car Design Project

Certification:	The participants, on successful completion of above trainings, will receive joint certification by BOSCH India & AKGEC.	
Program Coordinator:	Name: Mr. Vikram Singh Rajput	Contact number: 8851805766
Payment Mode:		